Solution Requirements

Date	02 July 2026
Team ID	xxxxxx
Project Name	A Comprehensive Measure of Well-Being (HDI Prediction System)
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	**Authentication**	Users (policymakers, researchers, students) must be able to securely register and log in using email/password or OAuth (Google/GitHub). Passwords must be securely hashed before storage.	High
2	**Authorization levels**	Define user roles: **Admin** (manages datasets, retrains models, manages users), **Researcher/Policymaker** (access to advanced analytics, batch uploads, and prediction history), and **Student/Guest** (basic single-country prediction access).	High

3	**External interfaces**	The system must integrate with the **UNDP Human Development Reports API** or **World Bank Open Data API** to automatically fetch and update baseline country statistics (Life Expectancy, Education, GNI).	Medium
4	**Transactions processing**	The system must process user-submitted country indicators, execute the Machine Learning inference pipeline, and log the transaction (prediction result) into the user's history database without data loss or duplication.	High
5	**Reporting**	The system must generate visual dashboards (using Matplotlib/Seaborn) showing the predicted HDI tier, comparative analysis with global averages, and allow users to export results as PDF or CSV files.	High
6	**Business rules, etc.**	The ML model must classify countries into four tiers based on standard UNDP thresholds: **Very High (≥0.800)**, **High (≥0.700 to <0.800)**, **Medium (≥0.550 to <0.700)**, and **Low (<0.550)**. Input validation must enforce realistic ranges (e.g., Life Expectancy 0-100, GNI > 0).	High
7	**Compliance to laws or regulations**	The system must comply with general data protection regulations (e.g., GDPR) regarding the secure storage, processing, and right-to-deletion of user Personally Identifiable Information (PII) like emails and names.	Medium

8	**Other**	The system must support **batch predictions**, allowing users to upload a CSV file containing multiple countries' data for bulk processing and downloading the results.	Low

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Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	**Performance & Speed**	The ML prediction inference and web page rendering must be highly responsive to ensure a smooth user experience without noticeable lag.	ML prediction inference time **< 1 second**. Web page load time **< 2 seconds**.
2	**Scalability**	The system architecture must handle growth in concurrent users and expanding historical datasets (adding more years and new countries) without performance degradation.	Supports up to **500 concurrent users**. Handles datasets up to **50 years of data for 190+ countries**.
3	**Security & Data Privacy**	All user data and API endpoints must be protected against common web vulnerabilities (SQL injection, XSS). Data in transit must be encrypted.	Passwords hashed using **bcrypt**. Data transmitted over **TLS 1.2/1.3**. ORM used to prevent SQL injection.
4	**Reliability & Availability**	The application must be highly available and include fault tolerance. If the primary ML model fails, the system should gracefully degrade or show a user-friendly error.	**99.5% uptime** per month. Automated fallback mechanism with **< 5% error rate** on valid inputs.

5	**Usability & Accessibility**	The user interface must be intuitive for non-technical users (like policymakers) and accessible to users with disabilities.	Responsive UI (mobile/desktop). Complies with **WCAG 2.1 AA standards** for color contrast and screen readers.
6	**Other (Maintainability & Portability)**	The codebase must be modular, well-documented, and easily deployable across different environments (local, cloud) without complex configuration	Code follows **PEP-8 standards**. System is **containerized using Docker** for seamless deployment on AWS/Azure/Heroku.